VANISHING OR SPLITTING: A QUOTIENT OF THE DUALIZING SHEAF WITH NO FIRST COHOMOLOGY SPLITS OFF UNLESS IT KILLS H^1

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the duality setup with the dimension count, the H^1 isomorphism under the vanishing hypothesis, the retraction from the universal property, and the identity check with the splitting conclusion are all retained verbatim.

Editorial note. The label lemma-vanishing names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Serre duality on the base field. Same chapter, Section section-Riemann-Roch. Gives dim H^1(X, omega_X) = 1 from H^0(X, O_X) = k.
- Representability of the H^1-dual functor. Duality for Schemes, Lemma duality-lemma-dualizing-module-proper-over-A. Makes omega_X represent F -> Hom_k(H^1(X, F), k) on coherent modules, the universal property used twice in the argument.
- Vanishing for zero-dimensional support. Cohomology of Schemes, Lemma coherent-lemma-coherent-support-dimension-0. Justifies the parenthetical H^1(X, Q) = 0 when Q is supported in dimension 0.

Source reference. Original lemma 0B3J: sources/stacks-chow/tags.txt, source lines 13967–13967. Prerequisite pointer invoked at proof line(s) 1334; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AWP: sources/stacks-chow/tags.txt, source lines 13727–13727. Prerequisite pointer invoked at proof line(s) 1342; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0B5B: sources/stacks-curves/pinned-original-source.tex, source lines 1109–1109. Prerequisite pointer invoked at proof line(s) 1336; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B5D: sources/stacks-curves/pinned-original-source.tex, source lines 1307–1307. Prerequisite pointer invoked at proof line(s)

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1322; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 0B5D. Given an exact sequence*

$$\omega_X \to \mathcal{F} \to \mathcal{Q} \to 0$$

of coherent $\mathcal{O}_X$-modules with $H^1(X, \mathcal{Q}) = 0$ (for example if $\dim(\mathit{Supp}(\mathcal{Q})) = 0$), then either $H^1(X, \mathcal{F}) = 0$ or $\mathcal{F} = \omega_X \oplus \mathcal{Q}$.

Proof. (The parenthetical statement follows from Cohomology of Schemes, Lemma 0B3J.) Since $H^0(X, \mathcal{O}_X) = k$ is dual to $H^1(X, \omega_X)$ (see Section 0B5B) we see that $\dim H^1(X, \omega_X) = 1$. The sheaf ω_X represents the functor $\mathcal{F} \mapsto \mathrm{Hom}_k(H^1(X, \mathcal{F}), k)$ on the category of coherent $\mathcal{O}_X$-modules (Duality for Schemes, Lemma 0AWP). Consider an exact sequence as in the statement of the lemma and assume that $H^1(X, \mathcal{F}) \neq 0$. Since $H^1(X, \mathcal{Q}) = 0$ we see that $H^1(X, \omega_X) \to H^1(X, \mathcal{F})$ is an isomorphism. By the universal property of ω_X stated above, we conclude there is a map $\mathcal{F} \to \omega_X$ whose action on H^1 is the inverse of this isomorphism. The composition $\omega_X \to \mathcal{F} \to \omega_X$ is the identity (by the universal property) and the lemma is proved. □